

1. Importance of Operation Research (OR)

Operation Research is important because it helps in **scientific and systematic decision-making**. Instead of guessing, OR uses **mathematical models, statistics, and optimization techniques** to find the best possible solution to real-life problems. It improves efficiency, reduces cost, saves time, and supports better planning and management in organizations. OR is widely used in industries where complex decisions must be made.

2. Uses / Applications of Operation Research

1. Business & Industry

- Minimizing production cost
- Maximizing profit
- Optimal resource allocation
- Production planning and scheduling

2. Transportation & Logistics

- Route optimization
- Vehicle scheduling
- Transportation cost minimization
- Airline and railway time-table planning

3. Manufacturing

- Inventory control
- Quality control
- Plant layout and machine scheduling

4. Military & Defense

- Weapon allocation
- Strategy making using game theory
- Logistics and supply chain management

5. Healthcare

- Patient scheduling
- Optimal use of hospital beds
- Shortening waiting time in OPD (queueing theory)

6. Finance & Banking

- Investment portfolio optimization
- Risk analysis
- Cash management

7. Government & Public Sector

- Urban planning
- Traffic control
- Resource distribution
- Disaster management

💡 Short Summary (for 3–4 marks)

Operation Research is important because it provides a **scientific approach** to decision-making. It helps organizations optimize their resources, reduce costs, improve efficiency, and solve complex problems. OR is used in **transportation, manufacturing, healthcare, finance, military strategy, inventory management, and project planning (PERT/CPM)**.